# llmXive Automated Review of Qwen-Image-VAE-2.0 Technical Report

*llmXive automated review panel*

**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence.

**1. "First f16 autoencoder" Claim (sec/experiment.tex, Sec 4.1.2):** The paper claims: "To the best of our knowledge, this is the first f16 autoencoder to achieve text fidelity exceeding f8 VAEs."

- **Evidence Check:** Table 2 lists `FLUX.2-dev` (f16c128) with an NED of 0.9535. The proposed `Qwen-Image-VAE-2.0-f16c128` has an NED of 0.9617.
- **Issue:** The claim is technically accurate regarding the *specific* f8 baselines listed (FLUX.1-dev is 0.9546). However, the phrasing "first f16 autoencoder" implies a broader survey. The table includes `FLUX.2-dev` (f16c128) which is an f16 model. While 0.9617 > 0.9535, the text does not explicitly state that `FLUX.2-dev` was evaluated on the *same* OmniDoc-TokenBench or if the comparison is strictly against the *f8* column. If `FLUX.2-dev` is a valid f16 competitor, the claim "exceeding f8 VAEs" is true, but the "first" claim is risky if other unlisted f16 models exist. The text should clarify the scope of the comparison to support the "first" assertion accurately.

**2. DINOv2 vs. DINOv3 Selection (sec/training.tex, Sec 3.2):** The paper claims: "Through extensive ablation studies... we find that DINOv2 consistently outperforms other candidates... (including DINOv3...)."

- **Evidence Check:** The bibliography includes `dinov3` (2025). The text asserts DINOv2 is superior.
- **Issue:** While the authors state they performed ablations, the paper does not present the specific results (e.g., a table or figure) comparing DINOv2 vs. DINOv3. Given that DINOv3 is a newer model (2025), the claim that it is *worse* than DINOv2 for this specific task is a strong factual assertion that requires explicit evidence in the text or a reference to a specific ablation table (which is missing). Without this, the claim relies entirely on the authors' word.

**3. Channel Dimension vs. Compression Ratio in f32 Comparison (sec/experiment.tex, Sec 4.1.1):** The paper claims: "our f32c192 achieves 0.8908 SSIM... outperforming all f32 competitors by substantial margins."

- **Evidence Check:** Table 1 shows `Qwen-Image-VAE-2.0-f32c192` (SSIM 0.8908) vs. `LTX-2` (f32c128, SSIM 0.7925) and `LTX-Video` (f32c128, SSIM 0.8329).
- **Issue:** The proposed model has 192 channels, while the listed f32 baselines have 32, 64, or 128 channels. The "substantial margin" might be partly due to the increased channel dimension (192 vs 128) rather than the architecture or compression ratio alone. The text attributes the success to "refined VAE architecture, expanded channel dimensions," but the comparison in the sentence "outperforming all f32 competitors" conflates the effect of the channel expansion with the compression ratio. A more accurate claim would specify "outperforming f32 baselines with lower channel dimensions" or acknowledge that the channel count is a variable.

**Conclusion:** The paper's core claims are generally supported by the data, but there are minor instances where the strength of the claim ("first", "consistently outperforms") slightly exceeds the explicit evidence presented in the text. These are writing/precision issues rather than fatal scientific errors.

**Action items.**

- [**writing**] In sec/experiment.tex, the claim 'first f16 autoencoder to exceed f8 VAEs' is ambiguous. Table 2 shows FLUX.2-dev (f16) with NED 0.9535, lower than the proposed 0.9617. Clarify if this comparison includes all f16 models or only specific baselines to support the 'first' claim accurately.
- [**writing**] In sec/training.tex, the claim that DINOv2 'consistently outperforms' DINOv3 lacks supporting data. The bibliography lists DINOv3 (2025), but no ablation results comparing the two are shown. Provide specific metrics or a reference to the ablation table to substantiate this selection.
- [**writing**] In sec/experiment.tex, the claim that f32c192 outperforms 'all f32 competitors' conflates channel dimension with compression ratio. Baselines like LTX-2 use 128 channels, while the proposed model uses 192. Clarify if the margin is due to architecture or the higher channel count to ensure accurate attribution.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure effectively visualizes the dataset categories but suffers from a caption typo, a label spelling inconsistency, and poor legibility in the Financial Report panel.

### Figure 2

The figure effectively demonstrates the visual superiority of the proposed VAE over baselines in preserving text clarity, but the column labels in the bottom row are significantly misaligned with the image panels, and the caption contains formatting errors.

### Figure 3

The figure displays high-quality image samples but fails to align with the caption's claim of using the ImageNet dataset, as the subjects appear to be from a different dataset (likely COCO). Additionally, the technical parameters labeling the rows (f16c64, etc.) are undefined within the figure or caption, hindering the ability to interpret the comparison.

Figure 4

Figure 4 effectively compares the three architectures visually, but the reconstruction loss title lacks parameter definition and the PSNR plot inconsistently displays uncertainty shading across models.

**Action items.**

- [**writing**] Figure 1: The caption contains a rendering artifact '$$3K' which should be corrected to '3K'.
- [**science**] Figure 1: The central label 'OmniDoc Token Bench' is misspelled (missing hyphen) compared to the caption 'OmniDoc-TokenBench'.
- [**writing**] Figure 1: The 'Financial Report' panel contains illegible text due to low resolution and excessive compression artifacts.
- [**fatal**] Figure 2: The rendered image is a composite of 11 distinct panels (Cosmos-0.1-CI16x16 through Original) but lacks a single unifying caption or title within the image frame to identify it as Figure 2.
- [**science**] Figure 2: The bottom row labels are misaligned with the image columns; specifically, the 'Ours-f16c64' label is positioned under the 'Stepvideo-T2V' column, and the 'Ours-f16c128' label is under the 'FLUX.2-dev' column, making it impossible to identify the correct baseline for comparison.
- [**writing**] Figure 2: The caption contains raw formatting artifacts ('[RGB]215,36,36') that should be removed or formatted as a color code for readability.
- [**science**] Figure 3: The caption claims samples are generated by SiT on ImageNet, but the images shown (eagle, dog, goldfish, tiger, panda, daisy, apples, burger, baseball, mountain) do not correspond to the standard ImageNet-1k class set (e.g., 'golden retriever' is not a class, 'goldfish' is 'goldfish' but the specific breeds and compositions look like COCO or custom datasets). This suggests a mismatch between the claimed dataset and the visual evidence.
- [**writing**] Figure 3: The labels 'Qwen-Image-VAE-2.0-f16c64', 'f16c128', 'f32c128', 'f32c192' and resolutions '256 x 256', '512 x 512' are placed as headers/footers for rows but lack a clear legend or axis definition explaining what 'f' and 'c' parameters represent (e.g., latent channels, compression factors), making the comparison difficult to interpret without external context.
- [**writing**] Figure 4: The title 'Reconstruction Loss (Smoothed0.9)' is ambiguous; the caption does not specify the smoothing parameter (e.g., 0.9) or the method used, making the metric definition incomplete.
- [**science**] Figure 4: The PSNR Performance plot shows a filled shaded region for the GSC (purple) curve but not for NSC or LSC, implying uncertainty or variance data is only presented for one model without explanation in the caption or legend.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of specialized terminology and acronyms that are often introduced without sufficient definition for a broad audience. In the Abstract, the term "diffusability" is defined only in a footnote, which is easily missed; this concept is central to the paper's contribution and should be explicitly defined in the main text flow. Similarly, acronyms such as "GSC" (Global Skip Connections), "DiT" (Diffusion Transformer), and "NED" (Normalized Edit Distance) are deployed as if they are common knowledge. While "VAE" is standard, "DiT" and "NED" are not universal enough to skip definition.

In Section 2.1 (OmniDoc-TokenBench), the term "logographic" is used to describe Chinese text. While accurate, it may be opaque to readers without a linguistics background; a brief clarification (e.g., "logographic (character-based) text") would improve accessibility. Furthermore, the Abstract introduces "OmniDoc-TokenBench" and "NED" in close proximity without defining the latter, creating a barrier to understanding the evaluation methodology.

The Introduction and Model sections rely heavily on shorthand like "f16" and "f32" for compression ratios. While these are defined in the Model section, they appear in the Introduction and Conclusion without immediate context, potentially confusing readers who do not track the variable definitions. The paper would benefit from a consistent policy of defining every acronym and specialized term (e.g., "semantic alignment," "latent space") at their first occurrence in each major section or at least in the Abstract and Introduction. The current reliance on footnotes and implicit knowledge reduces the paper's accessibility to non-specialists in the specific sub-field of high-compression VAEs.

**Action items.**

- [**writing**] Define 'diffusability' at first use in the Abstract. The current footnote is insufficient for a general audience; integrate the definition into the main text or provide a clearer, non-footnote explanation.
- [**writing**] Replace the acronym 'GSC' with 'Global Skip Connections' in full upon first mention in the Abstract and Introduction. Acronyms should be defined before use.
- [**writing**] Define 'DiT' (Diffusion Transformer) at its first occurrence in the Introduction. The term is used as a standard noun without prior expansion, excluding non-specialist readers.
- [**writing**] Replace the term 'logographic' in the Abstract and Section 2.1 with 'character-based' or 'ideographic' if 'logographic' is not strictly necessary, or provide a brief parenthetical explanation (e.g., 'logographic (e.g., Chinese)') to aid non-linguists.
- [**writing**] Define 'NED' (Normalized Edit Distance) in the Abstract or immediately upon first use in Section 2.1. The acronym is used as a primary metric without initial expansion.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent narrative regarding the trade-offs between compression, reconstruction, and diffusability. However, there are specific logical gaps in the justification of methodological choices and the interpretation of experimental constraints.

First, in **sec/training.tex**, the authors justify removing the KL loss by stating that "target semantic features are not necessarily Gaussian-distributed." While this premise is likely true, the logical leap to the specific solution (using a cosine similarity loss) is not fully explained. If the goal is to avoid forcing a Gaussian prior, one might expect a different alignment mechanism. The current argument implies that KL loss is the *only* thing preventing non-Gaussian alignment, but the paper does not explicitly rule out other non-Gaussian priors or explain why cosine similarity is the logical consequence of non-Gaussian targets. This creates a slight disconnect between the problem statement and the proposed solution.

Second, in **sec/experiment.tex**, the authors claim that "higher-dimensional latent space often require larger optimal Classifier-Free Guidance (CFG) scales" but then report results "without guidance" to ensure a "fair comparison." This reasoning contains a potential logical flaw. If high-dimensional latents *require* CFG to function well (i.e., to stabilize generation or achieve good FID), then evaluating them *without* CFG might artificially suppress their performance relative to lower-dimensional baselines that do not require CFG as strongly. By removing the very mechanism (CFG) that the authors claim is necessary for high-dimensional spaces, they may be introducing a bias that invalidates the conclusion that their high-dimensional models have "superior diffusability." The logic should be that they are testing the *intrinsic* quality of the latent space, but the justification provided ("to ensure fair comparison") contradicts the premise that CFG is needed for these specific models.

Third, in **sec/model.tex**, the authors argue that increasing the channel dimension $C$ does not affect DiT training efficiency because the DiT projects latents to a fixed hidden dimension. While the *sequence length* remains constant, the computational cost of the initial linear projection layer is $O(C \times H_{hidden})$. If $C$ triples (e.g., from 64 to 192), the FLOPs for this specific layer triple. The claim that complexity remains "nearly invariant" is an overstatement; it is invariant regarding the *transformer blocks* (which depend on sequence length), but not the *input projection*. This distinction is important for a technical report claiming efficiency.

These issues do not invalidate the core results but require clarification to ensure the logical chain from premise to conclusion is unbroken.

**Action items.**

- [**writing**] The paper presents a coherent narrative regarding the trade-offs between compression, reconstruction, and diffusability. However, there are specific logical gaps in the justification of methodological choices and the interpretation of experimental constraints.

First, in sec/training.tex, the authors justify removing the KL loss by stating that "target semantic features are not necessarily Gaussian-distributed." While this premise is likely true, the logical leap to the specific solution (using a

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the novelty and superiority of Qwen-Image-VAE-2.0, particularly in the Abstract, Introduction, and Conclusion sections. While the empirical results are impressive, the language used often extrapolates beyond what the specific data points justify, risking over-claiming.

First, the assertion in Section 4.1.2 and the Abstract that this is the "first f16 autoencoder to achieve text fidelity exceeding f8 VAEs" is a significant overreach. Table 2 (sec/experiment.tex) shows that the proposed f16c128 model achieves an NED of 0.9617, while the f8c16 FLUX.1-dev baseline achieves 0.9546. The difference is marginal (0.0071). Without reporting statistical significance (e.g., p-values or confidence intervals) or demonstrating a consistent, large margin across multiple random seeds or sub-splits, declaring a definitive "first" based on such a narrow gap is scientifically unsound. The claim should be qualified to reflect that the model "approaches or slightly exceeds" the performance of leading f8 baselines, rather than definitively surpassing them as a new class leader.

Second, the Conclusion states that the work "demonstrates a clear technical path to resolving the fundamental tripartite trade-off." This phrasing suggests the problem is solved or that the model dominates all three axes simultaneously without compromise. However, the data shows trade-offs still exist; for instance, while the f16c128 model has high NED, the f16c128 FLUX.2-dev baseline has a lower FID (0.73 vs 0.79) in Table 2. The model does not strictly Pareto-dominate all competitors in every metric. The language should be adjusted to "mitigating" or "significantly improving the balance of" the trade-off, rather than "resolving" it.

Finally, the claim that the model "surpasses all evaluated f8 VAEs" in text fidelity is technically accurate for the specific f16c128 configuration but is presented in a way that obscures the closeness of the competition with FLUX.1-dev. The narrative implies a decisive victory where the data shows a very tight race. The authors should avoid absolute superlatives like "surpasses all" when the margin is within the likely noise floor of the evaluation metric (OCR-based NED), which can vary based on the specific OCR model version or random sampling.

**Action items.**

- [**science**] The claim in the Abstract and Conclusion that Qwen-Image-VAE-2.0 is the 'first f16 autoencoder to achieve text fidelity exceeding f8 VAEs' is an overreach. Table 2 shows FLUX.2-dev (f16c128) achieving NED 0.9535, which is extremely close to the proposed model's 0.9617. Without statistical significance testing or a clear margin of superiority, declaring a ‘first’ based on a 0.8% absolute difference is unsupported.

- **[writing]** The conclusion states the model ‘resolves the fundamental tripartite trade-off’ between compression, fidelity, and diffusability. This is an absolute claim not fully supported by the data. While the model performs well, it does not strictly dominate all baselines in all metrics simultaneously (e.g., FLUX.2-dev has a lower FID in Table 2). The language should be tempered to reflect ‘significant improvement’ rather than a complete resolution of the trade-off.
- **[writing]** The claim that the model ‘surpasses all evaluated f8 VAEs’ in text fidelity (Section 4.1.2) is technically true for the specific f16c128 variant but ignores the f8c16 FLUX.1-dev baseline which is very close (0.9546 vs 0.9617). The phrasing implies a more decisive victory than the data suggests, potentially misleading readers about the magnitude of the improvement over the strongest f8 competitor.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a technical report on Qwen-Image-VAE-2.0, focusing on high-compression image encoding. From a safety and ethics perspective, the primary concerns revolve around data provenance, privacy, and potential dual-use applications, particularly given the model’s enhanced capability in text-rich scenarios.

**Data Provenance and Intellectual Property:** In `sec/data.tex`, the authors mention curating a “specialized document corpus” including academic papers, posters, and web pages, and using a synthetic pipeline to render text. There is no explicit statement regarding the licensing or copyright status of these source materials. Given the high fidelity of the reconstruction (especially for text), there is a risk that the model could inadvertently memorize and reproduce copyrighted text or proprietary layouts. The authors should clarify the legal basis for using these documents and whether the synthetic pipeline uses licensed fonts or open-source alternatives.

**Privacy and PII:** The construction of `OmniDoc-TokenBench` described in `sec/bench.tex` involves processing real-world document images. While the paper mentions filtering for text density, it does not address the presence of Personally Identifiable Information (PII) such as names, addresses, or financial data that might be present in financial reports, exam papers, or personal notes. The authors must confirm that a privacy review was conducted and that any sensitive information was redacted or anonymized prior to inclusion in the benchmark or training set.

**Dual-Use and Misuse Potential:** The paper highlights state-of-the-art performance in reconstructing text at high compression ratios (e.g., `f32`), which is a significant advancement. However, this capability also lowers the barrier for generating high-quality forgeries of documents, certificates, or official records. In `sec/training.tex`, the authors note the removal of KL and GAN losses to improve performance. While this improves efficiency, it may also make the latent space more susceptible to manipulation for generating deceptive content. The authors should include a brief discussion on the potential for misuse (e.g., document forgery) and any intended safeguards or usage policies for the released model weights.

**Conclusion:** The paper is technically sound but requires clarification on data licensing, privacy handling, and dual-use risks before it can be fully accepted from an ethics standpoint. These are primarily writing-level additions to the manuscript's data and discussion sections.

**Action items.**

- [**writing**] The paper describes a synthetic rendering pipeline for text-rich images (sec/-data.tex) but lacks explicit disclosure regarding the use of copyrighted documents or proprietary fonts in the training data. Authors must clarify the licensing status of the source materials used for the 'specialized document corpus' and synthetic rendering to ensure no IP infringement or unauthorized data usage.
- [**writing**] The OmniDoc-TokenBench construction (sec/bench.tex) involves cropping and processing real-world document images. The authors must explicitly state whether these documents contain personally identifiable information (PII) or sensitive data, and confirm that a privacy review or anonymization protocol was applied before inclusion in the benchmark.
- [**writing**] The removal of KL regularization and GAN loss (sec/training.tex) creates a latent space optimized purely for reconstruction and semantic alignment. The authors should briefly discuss potential dual-use risks, specifically whether the high-fidelity text reconstruction capabilities could be exploited to generate convincing forgeries or deepfakes of official documents, and if any mitigation strategies are considered.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence presented for Qwen-Image-VAE-2.0 is generally robust in terms of scale and benchmark coverage, but specific statistical rigor and experimental controls require clarification to fully support the central claims.

**Statistical Rigor in Downstream Generation:** The claim of "superior diffusability" and "rapid DiT convergence" is primarily supported by the Inception Score (IS) and gFID values in Table 1 (`sec/experiment.tex`). However, the text does not specify whether these metrics are averaged over multiple random seeds or represent a single run. In generative modeling, gFID and IS can exhibit significant variance depending on initialization and sampling stochasticity. Without reporting standard deviations or results from at least 3 independent seeds, it is difficult to determine if the observed improvements (e.g., gFID 9.52 vs 10.61 for FLUX.2-dev) are statistically significant or within the noise margin of the evaluation protocol.

**Benchmark Validity and Systematic Bias:** The introduction of OmniDoc-TokenBench is a strong contribution, but the evaluation methodology relies entirely on PP-OCRv5 to generate the "ground truth" text strings for the Normalized Edit Distance (NED) calculation (Eq. 1, `sec/bench.tex`). While the authors argue that using the same OCR model for both original and reconstructed images cancels out systematic errors, this assumes the OCR model's error distribution is uniform across all document types and fonts. If the baseline models produce artifacts that specifically confuse PP-OCRv5 (e.g., specific stroke merging patterns) while the proposed model does not, the metric may overstate the proposed model's advantage. A sensitivity analysis using a second, distinct OCR engine (e.g., Tesseract or a different PaddleOCR version) would strengthen the evidence that the improvement is in actual visual fidelity rather than OCR model alignment.

**Ablation Study Controls:** The ablation study for Global Skip Connections (GSC) in Figure 2 (`sec/model.tex`) claims to validate the design by comparing NSC, LSC, and GSC. The caption notes these are "trained from scratch." However, the main results in Table 1 compare the final models against baselines that may have been trained on different data distributions or for different durations. To isolate the effect of GSC, the ablation should explicitly control for total training steps and data composition, ensuring that the performance gain is not an artifact of a more favorable training schedule or data mix in the GSC variant.

**Data Scale Transparency:** The paper mentions scaling to "billions of images" (`sec/data.tex`) but does not provide a precise count or a breakdown of the data composition (e.g., ratio of natural images vs. synthetic text). Given the strong performance on text-rich benchmarks, the specific contribution of the synthetic rendering pipeline versus the real-world document corpus is a critical variable. While not a fatal flaw, providing a more granular data breakdown would allow for a better assessment of the "data engineering" claim.

Overall, the evidence is compelling but requires standard statistical reporting (seeds/std) and a discussion of potential OCR-induced biases to fully substantiate the claims of state-of-the-art performance.

**Action items.**

- [**science**] The claim of 'superior diffusability' relies on downstream SiT training results (Table 1) but lacks statistical significance testing or multiple random seeds. Please report mean/std over at least 3 seeds to rule out variance-driven claims.
- [**science**] The OmniDoc-TokenBench evaluation (Sec 3.2) uses a single OCR model (PP-OCRv5) as the ground truth proxy. This introduces systematic bias if the OCR model fails on specific fonts or layouts present in the test set. Please discuss this limitation or provide a sensitivity analysis using a second OCR engine.
- [**science**] The ablation study for Global Skip Connections (Fig 2) is described as 'trained from scratch' on f16c64, but the main results table compares against baselines trained on different data scales. Ensure the ablation controls for data scale and training steps to isolate the architectural contribution.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the evaluation in `sec/experiment.tex` requires strengthening to support the strong claims of state-of-the-art performance.

**1. Lack of Variance Estimates:** Tables 1 (`tab:main_bench`) and 2 (`tab:text_bench`) present single point estimates for all metrics (PSNR, SSIM, NED, gFID, IS). In deep learning benchmarks, performance can fluctuate based on random seeds, data shuffling, and initialization. Without reporting standard deviations (std) or confidence intervals (CI) across multiple runs, it is statistically impossible to determine if the observed improvements (e.g., the 0.9617 NED for Qwen-Image-VAE-2.0-f16c128 vs. 0.9546 for FLUX.1-dev) are statistically significant or within the margin of error. The authors must report results averaged over at least 3 independent runs with standard deviations.

**2. Missing Significance Testing:** The paper claims "state-of-the-art" and "surpassing" baselines based on point estimates. No hypothesis testing (e.g., t-tests or Wilcoxon signed-rank tests) is performed to validate these claims. For instance, in the text rendering section (Section 4.1.3), the authors note that "Stepvideo-T2V achieves notably higher NED than HunyuanImage-3.0... despite modest SSIM differences." While this observation is qualitative, a formal statistical test is needed to confirm if the difference in NED is significant given the variance in the data.

**3. Correlation Analysis Methodology:** In Section 4.1.3, the authors discuss the "imperfect correlation" between pixel metrics and text fidelity. However, they do not quantify this relationship. A Pearson or Spearman correlation coefficient with a corresponding p-value should be calculated across the benchmark results to rigorously demonstrate the necessity of the NED metric. Currently, the argument relies on anecdotal examples rather than statistical evidence.

**4. Reproducibility of Statistical Claims:** The "Diffusability" evaluation (Section 4.1.3) mentions training SiT on ImageNet. The results (IS and gFID) are highly sensitive to the number of training steps and random seeds. The manuscript states results are reported at "80 epochs" but does not specify if these are the mean of multiple seeds. Given the high stakes of the "superior diffusability" claim, the absence of variance metrics undermines the reproducibility of the statistical conclusion.

The authors should re-run experiments with multiple seeds, report mean ± std for all tables, and include significance testing for key comparisons to meet the standards of statistical analysis required for a technical report of this magnitude.

**Action items.**

- [**science**] Report confidence intervals or standard deviations for all quantitative metrics (PSNR, SSIM, NED, gFID) in Tables 1 and 2. Currently, single point estimates are provided without measures of variance, making it impossible to assess statistical significance of the reported improvements over baselines.

- **[science]** Clarify the statistical methodology for the 'Correlation Analysis' in Section 4.1.3. The text notes discrepancies between pixel metrics and NED but does not provide a formal correlation coefficient (e.g., Pearson/Spearman) or p-value to quantify the strength and significance of this relationship.
- **[science]** Specify the number of random seeds or independent runs used to generate the results in Table 1 and Table 2. If results are from a single run, this must be explicitly stated as a limitation, as deep learning metrics can vary significantly based on initialization and data shuffling.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a technically dense and ambitious report on Qwen-Image-VAE-2.0. The overall structure is logical, moving from motivation to architecture, data, training, and evaluation. The writing is generally clear and professional, effectively communicating complex architectural choices and experimental results. However, there are several specific grammatical errors and syntactic issues that disrupt the flow and require correction before publication.

In the **Training** section (sec/training.tex), the authors state: "As the training progresses, we gradually loose the alignment margins." The word "loose" is an adjective meaning "not tight," whereas the verb form required here is "loosen." This is a clear grammatical error that should be fixed to "loosen."

In the **Experiments** section (sec/experiment.tex), under "Performance of Text Rendering," a sentence reads: "We compute NED on raw OCR outputs without text normalization, while minor spacing artifacts from OCR may inflate edit distance, the evaluation pipeline is applied consistently across all models ensuring fair comparison." This is a comma splice, joining two independent clauses ("We compute..." and "the evaluation pipeline is...") with only a comma and a subordinating conjunction used incorrectly. It should be restructured, for example: "While minor spacing artifacts from OCR may inflate edit distance, the evaluation pipeline is applied consistently..." or by using a semicolon.

Additionally, in the caption for Figure 4 (sec/experiment.tex), the text reads: "Qualitative comparison of text reconstruction on Ours OmniDoc-TokenBench." The use of the possessive pronoun "Ours" to modify the noun phrase "OmniDoc-TokenBench" is incorrect. It should be "our OmniDoc-TokenBench."

Finally, in the **Introduction** (sec/introduction.tex), the list of contributions includes "improved architecture, comprehensive data engineering, and enhanced training strategy." To maintain parallel structure, "strategy" should likely be pluralized to "strategies" to align with the plural nature of the other items or the context of multiple techniques described.

Addressing these specific points will significantly improve the readability and polish of the manuscript. The scientific content appears robust, but these linguistic refinements are necessary for a professional presentation.

**Action items.**

- **[writing]** In sec/training.tex, the phrase ‘gradually loose the alignment margins’ contains a grammatical error. ‘Loose’ is an adjective; the verb form required here is ‘loosen’. This should be corrected to ‘gradually loosen the alignment margins’.
- **[writing]** In sec/experiment.tex, the sentence ‘While minor spacing artifacts from OCR may inflate edit distance, the evaluation pipeline is applied consistently...’ is a comma splice. It joins two independent clauses with only a comma. Please insert a semicolon, a period, or a conjunction (e.g., ‘...edit distance; however, the evaluation...’).
- **[writing]** In sec/experiment.tex, the phrase ‘Ours OmniDoc-TokenBench’ in the caption of Figure 4 is grammatically incorrect. ‘Ours’ is a possessive pronoun and cannot modify a noun directly. It should be changed to ‘our OmniDoc-TokenBench’.
- **[writing]** In sec/introduction.tex, the phrase ‘designed to overcome these challenges through improved architecture, comprehensive data engineering, and enhanced training strategy’ lacks parallelism. ‘Strategy’ should be pluralized to ‘strategies’ to match ‘architecture’ and ‘engineering’ (or ‘engineering’ should be singular if ‘strategy’ is singular, but ‘strategies’ fits the context of multiple techniques better).